

LECTURE 2.3



Bias and Precision

ECONXXXXX Econometrics I

Sam Deegan
University College Dublin

PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

The Third of Four Lectures on One Estimator

Lectures already given are greyed and today's is marked. Lecture 2.1 fitted the line and 2.2 asked how much of earnings it accounts for. Today asks how far the fitted slope lands from the truth, and 2.4 asks how to report it.

What Lectures 1.3 and 2.2 Left Open

Lecture 1.3 asked three questions of the sample mean: is the rule unbiased, how far does it vary across samples, and what interval does it support.

Lecture 2.2 measured how much of the variation in earnings our line accounts for, on the sample it was fitted on and on the hold-out.

Neither says how far our 1.5227 sits from the population slope. Today asks 1.3's first two questions again, of a slope instead of a mean.

Question

If I drew a different 120 people from the same population and fitted again, would I get the same slope?
Should you want me to?

Lecture Plan

1 The Gauss-Markov Assumptions.

Five claims about the population, and what each one is needed for.

2 Unbiasedness.

The fitted slope is the true slope plus one term, and that term averages to zero.

3 Precision.

How far the slope varies across samples, and the three things that narrow it.

SECTION 1

The Gauss-Markov Assumptions

1. The Gauss-Markov Assumptions
2. Unbiasedness
3. Precision

Five Claims About the Population, Not the Sample

The arithmetic of 2.1 and 2.2 is true of any 120 rows, including rows of nonsense. Five assumptions turn it into a claim about the world:

- 1. Linear in Parameters.** In the population, $y = \beta_0 + \beta_1(x - 13) + u$.
- 2. Random Sampling.** Our 120 people were drawn at random from that population.
- 3. Variation in Schooling.** The 120 do not all have the same years of schooling.
- 4. Zero Conditional Mean.** At every level of schooling, the error averages to zero.
- 5. Constant Error Variance.** At every level of schooling, the error has the same variance.

Wooldridge Numbers These SLR.1 to SLR.5

SLR is simple linear regression. Lecture 3.1 restates all five for more than one regressor and rennumbers them MLR.1 to MLR.5. The five together are named after the theorem in today's third section (Wooldridge, 2020).

Assumption Four: Zero Conditional Mean

Assumption four is the one that is almost never true outside a simulation, and it is the one this module returns to most often:

$$E[u \mid x] = 0 \quad (1)$$

- u : everything other than schooling that raises or lowers earnings
- $E[u \mid x]$: the average of that, among people with a given schooling

Lecture 2.1 used this same condition for a different purpose: it is what makes the line equal to the conditional expectation. Here it has a second job, and it is the only assumption that makes the fitted slope unbiased.

Assumption Four Is a Claim About Ability

It says people who stayed longer in school are no different in anything else that raises pay. In this population it holds exactly, because the generator draws the error independently of schooling. In Irish earnings data it would not, and Lecture 3.1 puts ability into the model.

Assumption Five: Constant Error Variance

Assumption five is about the spread of the error, and says nothing at all about its average:

$$\text{Var}(u \mid x) = \sigma^2 \quad (2)$$

- σ^2 : the variance of the error, the same number at every level of schooling
- σ : its standard deviation, 4.50 in this population

It is not needed for unbiasedness, and Wooldridge says so explicitly. It is added because it turns a long variance expression into one short formula, and because the theorem in section 3 is false without it (Wooldridge, 2020).

Where This One Fails, and What Fails With It

If earnings spread out more at higher levels of schooling, assumption five fails. The estimate stays unbiased and the standard error stops being right. Lecture 4.1 turns that on and repairs the standard error.

What Each Assumption Is Needed For

No two of the five deliver the same result, and only one of them delivers unbiasedness:

- **One** sets the shape, and Lecture 4.1 shows what a curved population does to a line.
- **Two** is how the data arrived, and it is the one nobody checks.
- **Three** keeps S_{xx} away from zero, so that a slope exists to compute.
- **Four** is what makes the estimator unbiased, and nothing else delivers that.
- **Five** is what makes the variance formula in section 3 correct.

Four and Five Fail in Different Ways

Assumption four is about the average of the error and five is about its spread. Break four and the estimate is centred in the wrong place. Break five and it is centred correctly with the wrong standard error printed beside it.

SECTION 2

Unbiasedness

1. The Gauss-Markov Assumptions
2. Unbiasedness
3. Precision

One Sample Fitted Against the Line We Know Is True

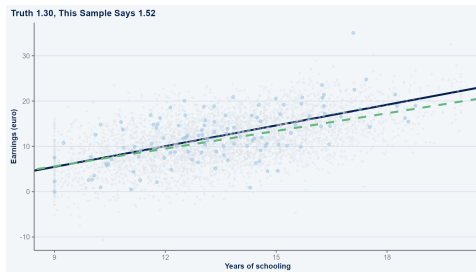


Figure 1: Grey is the population, colour our 120, dashed the true line, solid our fitted one.

Two Lines, and Only One of Them Is Ever Available

The lines differ because our 120 people are one draw. No dataset shows both: the dashed line is on this slide only because we wrote the population ourselves.

The Slope Is a Weighted Sum of the Earnings

Before asking where the slope lands on average, rewrite 2.1's formula so that the only random part of it is visible:

$$\hat{\beta}_1 = \sum_{i=1}^n w_i y_i \quad w_i = \frac{x_i - \bar{x}}{S_{xx}} \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \quad (3)$$

- w_i : the weight person i carries, fixed once schooling is known
- S_{xx} : the sum of squared deviations of schooling, 633.9 in our sample

This is 2.1's covariance over variance and nothing else. The numerator $\sum (x_i - \bar{x})(y_i - \bar{y})$ equals $\sum (x_i - \bar{x}) y_i$, because $\bar{y} \sum (x_i - \bar{x})$ is zero (Wooldridge, 2020).

Two Facts About the Weights

$\sum w_i = 0$, because the deviations of schooling sum to zero. And $\sum w_i(x_i - 13) = 1$, because that numerator is S_{xx} itself. The next frame uses these two and nothing more; the appendix proves them.

The Slope Equals the True Slope Plus One Term

Substitute the population model for every y_i in that weighted sum, and apply the two facts about the weights:

$$\begin{aligned}\hat{\beta}_1 &= \sum_{i=1}^n w_i (\beta_0 + \beta_1(x_i - 13) + u_i) \\ &= \beta_0 \sum_{i=1}^n w_i + \beta_1 \sum_{i=1}^n w_i(x_i - 13) + \sum_{i=1}^n w_i u_i \\ &= \beta_1 + \sum_{i=1}^n w_i u_i\end{aligned}\tag{4}$$

The first sum is zero, so the intercept disappears. The second sum is one, so the true slope comes through unchanged. Everything else is the third term.

One Term Separates the Estimate From the Truth

$\sum w_i u_i$ is the entire distance between $\hat{\beta}_1$ and β_1 . It is a weighted sum of 120 errors nobody observes, 10

Why the Rule Lands on the True Slope on Average

The weights are built from schooling alone, so across repeated samples only the errors vary. Take the average of the decomposition:

$$E[\hat{\beta}_1 | x] = \beta_1 + \sum_{i=1}^n w_i E[u_i | x] = \beta_1 \quad (5)$$

Assumption four sets every one of those conditional averages to zero. A weighted sum of zeros is zero, so the rule is centred on β_1 . It holds at any set of schooling values, so it holds without conditioning on them (Wooldridge, 2020).

Only Assumption Four Is Used at This Step

Assumptions one to three were used to write the decomposition down. Assumption four is the only one this step needs, and assumption five is not used anywhere in it. That is why breaking five leaves unbiasedness intact.

Five Hundred Samples, Five Hundred Slopes

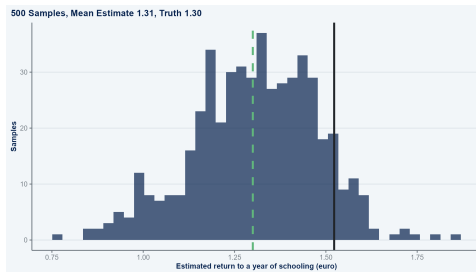


Figure 2: Five hundred samples of 120, one fitted slope kept from each, against the true 1.30.

The Histogram Is Centred on 1.30 and Our Draw Is Not

Each bar collects the samples whose fitted slope landed there. The dashed line is the true 1.30, the solid line our 1.5227. Lecture 1.3 drew this same picture for a mean, and only the quantity on the horizontal axis is different.

Unbiasedness Is a Property of the Rule, Not of Your Estimate

Our sample gave 1.5227 when the true slope is 1.30. We missed by 0.2227, and nothing went wrong in the arithmetic or in the sampling.

Unbiasedness is a statement about where that histogram sits. It says nothing about any single bar in it, and so it says nothing about ours (Wooldridge, 2020).

In real work you draw once and never see the histogram. Everything in Lecture 2.4 is about what one draw honestly supports.

An Unbiased Rule Still Misses in Almost Every Sample

What an unbiased rule does not do is miss in the same direction on average. Both facts hold at once: least squares is unbiased, and our 1.5227 is wrong. A student who reads unbiased as “correct” cannot be taught the rest of block 2.

Always Dropping the Highest Earner

A colleague fits every regression after deleting the highest-earning person in the sample. The deletion is decided before the data arrive, so by Lecture 1.3's definition this is an estimator, and it has a sampling distribution of its own.

Question

Is that estimator unbiased for β_1 , and which of the five assumptions does the deletion break?

SECTION 3

Precision

1. The Gauss-Markov Assumptions
2. Unbiasedness
3. Precision

Deriving the Variance of the Slope

Unbiasedness took the average of the leftover term. Precision takes its variance, and the weights are constants throughout:

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var}\left(\sum_{i=1}^n w_i u_i\right) = \sum_{i=1}^n w_i^2 \text{Var}(u_i) = \sigma^2 \sum_{i=1}^n w_i^2 \\ &= \sigma^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}\end{aligned}\tag{6}$$

The second step drops every cross term, because random sampling leaves different people's errors uncorrelated. The third step takes one σ^2 outside the sum, which is exactly what assumption five permits (Wooldridge, 2020).

Only the Errors Are Random in This Calculation

Each weight is built from schooling alone, so it comes out of the variance squared. Lecture 1.2's algebra of variances then applies line by line, and no rule beyond it is needed.

Three Ingredients, Read Off the Formula

One rewrite of the denominator makes the sample size visible, because $S_{xx} = (n - 1) s_x^2$:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} = \frac{\sigma^2}{(n - 1) s_x^2} \quad \text{sd}(\hat{\beta}_1) = \frac{\sigma}{s_x \sqrt{n - 1}} \quad (7)$$

- s_x^2 : the sample variance of schooling, computed in Lecture 1.1

Three ingredients, and each of them is visible in the formula rather than asserted beside it:

- **A larger error variance raises it.** σ is the only term on top.
- **More spread in schooling lowers it.** s_x is underneath, so a slope through people who barely differ is imprecise.
- **More people lower it.** $\sqrt{n - 1}$ is underneath, so four times the data halves the spread.

Only One of the Three Is Yours to Set

ECON is a fact about the world and s_x a fact about who was sampled, so n is usually the only one you control. 16

The Standard Error on Our 120 People

σ is a population number we never see, so the residuals from 2.2 replace the errors and the formula becomes computable:

$$\hat{\sigma} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{2692.8}{118}} = 4.777 \quad \text{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} = \frac{4.777}{\sqrt{633.9}} = 0.1897 \quad (8)$$

- $\hat{\sigma}$: the estimated standard deviation of the error
- $\text{se}(\hat{\beta}_1)$: the estimated standard deviation of the slope across samples

All three ingredients are inside that second fraction. The 4.777 is the error, and the 633.9 contains both the spread of schooling and the 120 people. It is the spread of the histogram, not the spread of earnings (Angrist & Pischke, 2015).

The Estimate of the Error Also Misses

The true error standard deviation is 4.50 and our residuals give 4.777. The divisor is $n - 2$ rather than n because the two fitted coefficients have already used up two of the 120 residuals, which the appendix

Four Times the Sample

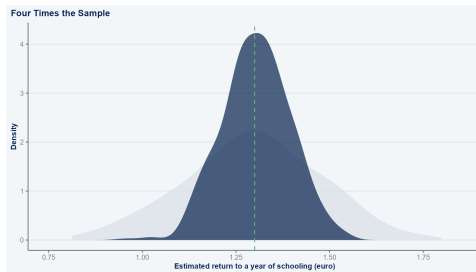


Figure 3: Five hundred samples at 480 people, drawn against the same exercise at 120.

Quadrupling the Sample Halves the Spread

Both densities are centred on 1.30, because sample size has nothing to do with unbiasedness. Only the width differs, and $\sqrt{n - 1}$ sets the rate at which it falls: four times the data, half the spread.

More Spread in Schooling

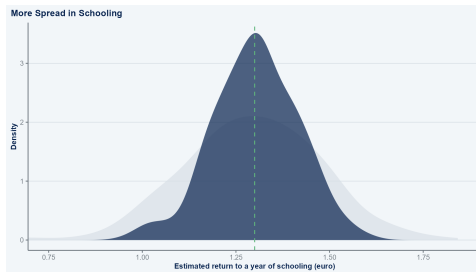


Figure 4: Five hundred samples from a population whose schooling is more spread out.

A Wider Range of Schooling Narrows the Distribution

Here s_x is larger, so S_{xx} is larger and the variance is smaller. If everyone in a sample left school within a year of everyone else, no amount of data estimates the slope precisely.

A Smaller Error Standard Deviation

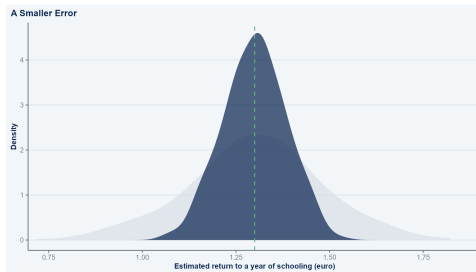


Figure 5: Five hundred samples from a population whose error standard deviation is smaller.

A Less Variable Error Narrows the Distribution

σ is the only ingredient on top of the formula, and it is a fact about the population rather than about the study. No sample size and no sampling design lowers it.

Two Hundred People, or Eighty

Two samples are offered and you may have only one. Two hundred people whose schooling runs from eleven to thirteen years, or eighty whose schooling runs from nine to twenty.

Question

Which of the two gives the more precise estimate of the slope, and which two ingredients have you traded against each other?

Least Squares Against a Two-Point Estimator

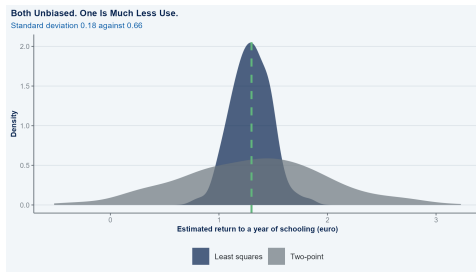


Figure 6: Least squares against a slope computed from the two extreme observations only.

Both Are Centred on 1.30, and One Is Far Wider

The two-point rule takes the lowest and the highest schooling in the sample and joins them. It is a weighted sum of the earnings, it is unbiased, and it throws away the other 118 people.

The Gauss-Markov Theorem

The two-point rule is not an isolated bad idea. Under all five assumptions, nothing in its class beats least squares, and each word of the claim is restrictive (Wooldridge, 2020):

- **Linear** means the estimator is a weighted sum of the y_i , the form derived above.
- **Unbiased** means centred on β_1 across samples, which the two-point rule also is.
- **Best** means smallest variance, and it means nothing beyond that.

Best Linear Unbiased

Among all estimators that are weighted sums of the earnings and unbiased for β_1 , least squares has the smallest sampling variance. This is where the name Gauss-Markov assumptions comes from.

What the Theorem Does Not Claim

Three limits, and the third is the one most often overlooked:

- It is a claim about a **class**. Step outside linear or outside unbiased and the theorem is silent.
- It rests on **all five** assumptions, and on the fifth in particular.
- It is about **precision**, not about being centred in the right place.

A Biased Rule Can Have a Smaller Variance

Lecture 1.3's trimmed mean was steadier than the sample mean and biased, so the theorem neither permits nor forbids it. Lecture 4.1 turns assumption five off: least squares stops being best and the standard error stops being right, which is where robust standard errors come from.

What Would Have To Be True?

This time the question is asked of the standard error we just computed, 0.1897, and of the claim that the rule producing 1.5227 is centred on the truth.

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Application

The app draws five hundred samples, fits each one, and builds the histogram of slopes, with the five assumptions switchable one at a time.

Question

If we break assumption four deliberately, does the histogram shift, widen, do both, or do neither?

The lab for this lecture is the simulation loop: `replicate()`, then `sd()` on the five hundred slopes, checked against the printed standard error.

Summary: Assumptions, Unbiasedness and Precision

1 The Gauss-Markov Assumptions.

Assumption four makes the estimator unbiased; assumption five makes the variance formula correct.

2 Unbiasedness.

The fitted slope is β_1 plus a weighted sum of errors, and assumption four sets that sum's average to zero.

3 Precision.

The same sum has variance σ^2/S_{xx} , which on our 120 people gives a standard error of 0.1897.

Next Lecture

Lecture 2.4 takes today's two numbers, 1.5227 and 0.1897, and turns them into something that can be written down and defended:

- **Testing a Coefficient:** comparing an estimate with its own standard error.
- **Confidence Intervals for an Effect:** a range instead of a point, and whether it covers 1.30.

Today's histogram is gone from 2.4 onwards. The standard error, 0.1897, is the only part of it that a single sample gives you.

Before Lecture 2.4

1 **Play:**

The app. Raise the sample size, then widen schooling, and see which narrows the histogram more.

2 **Apply:**

Problem set 2, covering 2.3 and 2.4. It repeats today's decomposition with $n = 6$ and numbers you can do by hand.

3 **Review:**

Wooldridge (2020), section 2-5, on the five assumptions, unbiasedness and the variance of the slope.

Thank you

sam.deegan@ucdconnect.ie
sam-deegan.com



BACKUP

Appendix

Derivation: The Two Facts About the Weights

The body frames used $\sum w_i = 0$ and $\sum w_i(x_i - 13) = 1$ without proving them. Both are summation algebra from Lecture 1.1:

$$\sum_{i=1}^n (x_i - \bar{x}) = 0 \qquad \sum_{i=1}^n (x_i - \bar{x})(x_i - 13) = S_{xx} \qquad (9)$$

The Centring Constant Cancels Once Again

The second holds because the first is zero: replacing $x_i - 13$ by $(x_i - \bar{x}) + (\bar{x} - 13)$ adds a multiple of the first sum, which contributes nothing. Dividing both by S_{xx} gives the two facts.

Derivation: Why the Divisor Is n Minus Two

The body frames used $\hat{\sigma}^2 = \text{RSS}/(n - 2)$. Replacing the unobserved errors by residuals costs two restrictions, the two first-order conditions from Lecture 2.1:

$$\sum_{i=1}^n \hat{u}_i = 0 \qquad \sum_{i=1}^n (x_i - 13) \hat{u}_i = 0 \qquad (10)$$

Two Conditions, Two Degrees of Freedom

Given 118 of the residuals, those two equations return the other two, so the 120 residuals carry only 118 free numbers. Dividing by 118 rather than 120 is what makes $\hat{\sigma}^2$ unbiased for σ^2 (Wooldridge, 2020).

Notes and Tools

- **The App:** the sampling app, with the five assumptions switchable one at a time.
- **The Lecture in One Line:** `summary(lm(y ~ I(x - 13)))$coefficients[2, 2]`.
- **This Lecture's Lab:** `replicate()` and `sd()`, checked against `summary()`.

References

Angrist, J. D., & Pischke, J.-S. (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.